ATLAS Repair and Re-Audit

Universal Proof Horizons for a Single ATP

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Repaired version: Research Draft 0.3 (13 pages)

Re-audit verdict: PASS for internal mathematical coherence, subject to the explicit formal model and trust assumptions.

The substantive conclusions of the original draft survive the targeted repairs. In this re-audit, the earlier two high-priority findings (service accounting and canonical integer-code recognition) are *resolved in the revised written specification.* No new proof-invalidating defect was identified in the central constructions. This is a critical mathematical review and a successful LaTeX/PDF production check, *not* a mechanized proof, an independent soundness proof for an ATP implementation, or a guarantee that every conceivable error has been excluded.

1. Traceable repair ledger

Finding	Repair in draft 0.3	Disposition
A1: service cost	Definition 2.1 now permits one charge for an *atomic verified inference* including its local check. Separately performed verification is charged separately. Theorem 8.1 and Section 9 explicitly use this convention; terminal recognition of a completed accepted proof incurs no extra event.	**PASS**
A2: code recognition	Definition 2.1 now requires total decoding Dec, a computable injective Enc and an exact decidable canonical-code membership test. Remark 4.3 supplies the corresponding total, effectively proper cost function on $\mathbb{N}$.	**PASS**
Stage accounting	Definition 6.1 distinguishes paid BANK maintenance from the newly charged services at each standardized restart. The remaining-cost horizon is not confused with cumulative historical spending.	**PASS**
Clarifications	Explicit promise-class uniformity; different notation for baseline versus canonical optimum; better description of metalevel search cost; precise Section 10 heading; opening pointer typo repaired.	**PASS**
Visual exposition	Three labelled black-and-white schematic diagrams added to illustrate bounded search, a finite horizon ladder, and the restarted proof path.	**PASS**

2. Mathematical re-audit by dependency

Assertion	Status	Independent recheck
Definition 2.1 / Lemma 2.3	**PASS**	Complete, terminating finite successor lists produce finitely many histories at every bounded service depth; the added prefix-closure premise resolves the path convention.

Assertion	Status	Independent recheck
Theorem 3.1	**PASS**	Uniform breadth-first search yields a partial computable least-depth accepted history. Finite enumeration of the first successful level makes the least-code tie-breaker effective. Domain claim is restricted to promised admissible indices.
Proposition 4.2	**PASS**	Canonical chains and accepted event histories are inverse computable descriptions, each extension having unit cost. Nothing is inferred about arbitrary padded proofs.
Remark 4.3	**PASS**	The repaired decoder makes the valid-code branch decidable; valid bounded-cost histories form an effectively enumerable finite set, while invalid bounded-binary-length integers form another. This proves total computability and effective properness of the displayed cost extension.
Theorem 5.2	**PASS**	A legal accepted history of cost at most m immediately bounds the minimum by m. Fixed charged overhead adds K_{admin} when stipulated.
Proposition 5.4	**PASS**	Both projection and the $5m$ comparison are witnessed by strong, explicit noninterference, filler-service, scheduling and terminal-recognition hypotheses; no unconditional fivefold improvement is claimed.
Proposition 6.3	**PASS**	A cost-nonincreasing successful-route image from stage k to $k+1$ directly supplies the monotonicity witness. BANK inclusion alone is insufficient, as shown by the forced-scan counterexample.
Theorem 7.2	**PASS**	Every stage has a successful history by persistence from stage zero. Hence uniformly applied partial breadth-first search terminates at every stage. Monotone natural numbers have finitely many strict decreases, each decidable from two terminating computations.
Corollary 7.3	**PASS**	Optional, certified tautology deposits can increase BANK size indefinitely without creating any zero-cost accepted target or reducing the one-service target route.
Theorem 8.1	**PASS**	Verified inference costs exactly one event in the stated example. The stage-k BANK update is computable by bounded simulation; a first halt at $t \geq 1$ creates the sole drop from $t-1$ to t. A putative stabilization bound N would decide whether the simulated program halts by time N.
Corollaries 8.2–8.3	**PASS**	The limiting horizon equals 1 precisely on halting instances, and 2 otherwise; pointwise identification of completed drops does not imply a uniform final-stage announcement.
Section 10 identity	**PASS**	The realized-cost gap splits algebraically into realization excess and policy-accessibility excess; both are nonnegative under the stated existence assumptions.

3. Focused sanity checks

The two-rule example $a \to \lambda \to T$ was also checked with a small, independently written breadth-first enumeration over states consisting of known formulas. It returned minimum costs 2 from $\{a\}$ and 1 from $\{a, \lambda\}$. Bounded test traces had the following values:

Case	h_0	h_1	h_2	h_3	h_4
Never halts	2	2	2	2	2
First halt at stage 1	2	1	1	1	1
First halt at stage 3	2	2	2	1	1

The exercise is consistent with the reduction and catches elementary indexing errors. It does *not* verify the reduction for every Turing program; that general conclusion follows from the written proof, conditional on the standard undecidability of halting.

4. Non-negotiable limits of the clearance

1. **Relative optimality only.** The horizon is minimum charged service length over the explicitly represented legal histories; it is not a representation-independent shortest mathematical proof or a guaranteed wall-clock-optimal ATP method.
2. **Trusted operational semantics.** Exhaustive minimum certificates are reproducible relative to the declared complete successor and acceptance routines. This audit has not established that such routines are complete against a separately specified implementation or real-world operational semantics.
3. **No five-role equivalence.** The one-ATP construction accepts proofs of T, not every metalogical result of a five-role AMLD. The $5m$ statement is a conditional scheduling construction.
4. **Restart, not total cost savings.** Proof-horizon drops after BANK growth do not reimburse the historical computation used to discover and verify deposits.
5. **No external formalization.** No Lean, Coq, Metamath, or independent prover certificate was produced for the manuscript as a whole; the small computational sanity check is not formal proof verification.

5. Final disposition

ATLAS re-audit status: REPAIRS ACCEPTED; INTERNAL MATHEMATICAL AUDIT PASSED, conditional on the stated assumptions.

The revised paper's central four results have coherent proofs in the promised effective, finitely branching, verifier-gated singleton-history setting. The corrected presentation now specifies the exact cost model needed for the $2 \to 1$ reduction and the decoder needed for the effectively proper code-cost extension. Its principal positive and negative conclusions remain intact. Further mechanized formalization or inspection of a concrete ATP implementation would constitute a separate, stronger certification undertaking.

Audit performed using the research draft 0.2 source manuscript, its earlier ATLAS audit, and the generated 0.3 LaTeX/PDF; the cited external articles were not independently re-reviewed. Original draft 0.2 was not overwritten. Nothing has been published to a public repository.